Classical theories of information science

# 信息+机器学习 之一

# Minimum Description Length Principle

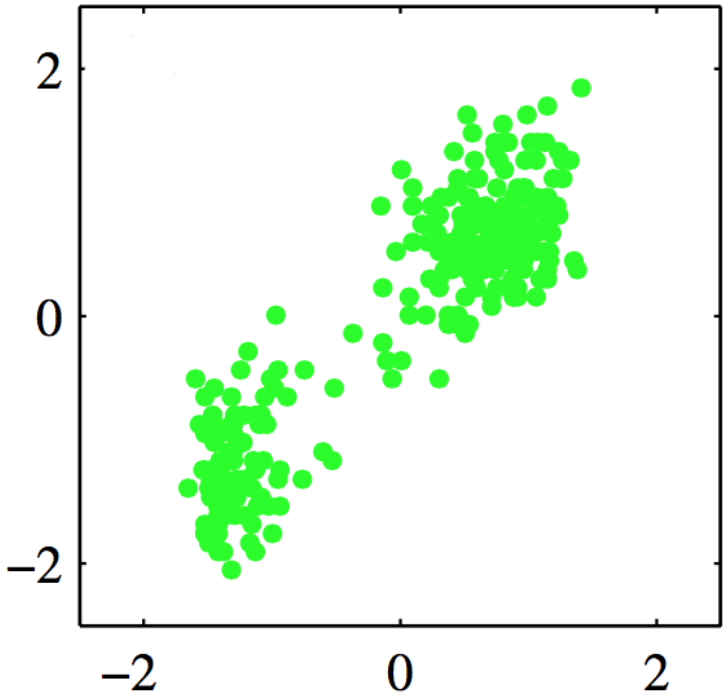
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# 1. Model Selection

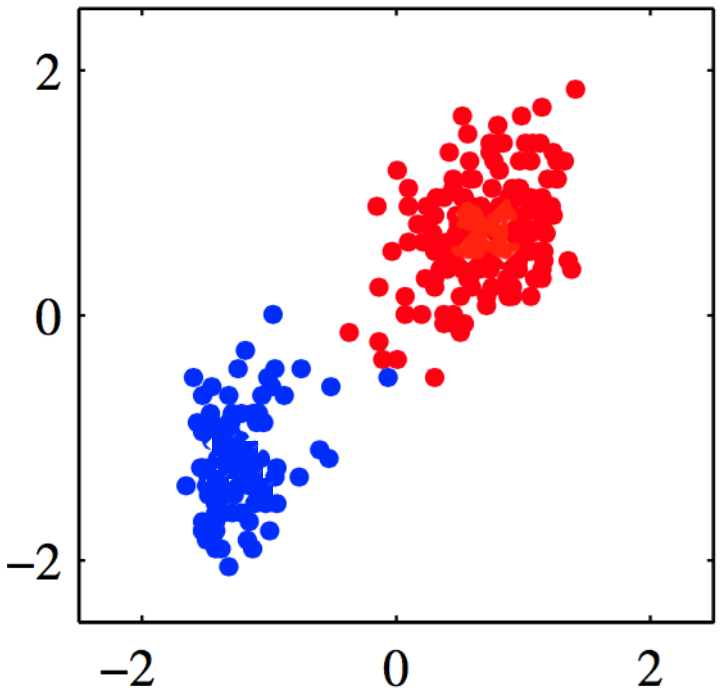
How does one decide among competing explanations of data given limited observations?

# The problem of clustering the data

We have the following data:




We want to cluster the data into two clusters (red and blue)




How?

# K-mean 算法

- 初始化均值点$\boldsymbol{\mu}_1,\ldots,\boldsymbol{\mu}_k$
- 迭代如下
  - 把每个数据点按照就近原则分配给相应的$\boldsymbol{\mu}_i$
  - 把$\boldsymbol{\mu}_i$更新为所分配的数据点的均值
- 迭代停止，如果聚类分配不变

Initialize $\boldsymbol{m}_i, i = 1, \ldots, k$, for example, to $k$ random $\boldsymbol{x}^t$

Repeat

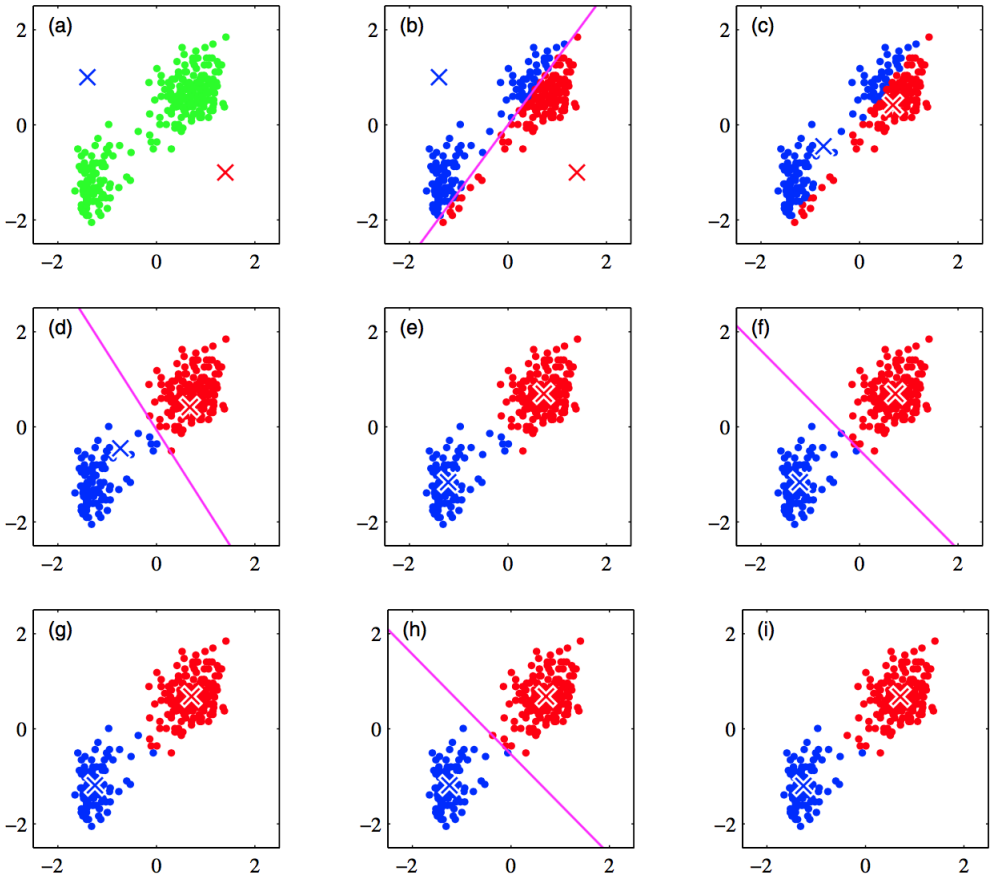
For all $\boldsymbol{x}^t \in \mathcal{X}$

$$b_i^t \leftarrow \begin{cases} 1 & \text{if } \|\boldsymbol{x}^t - \boldsymbol{m}_i\| = \min_j \|\boldsymbol{x}^t - \boldsymbol{m}_j\| \\ 0 & \text{otherwise} \end{cases}$$

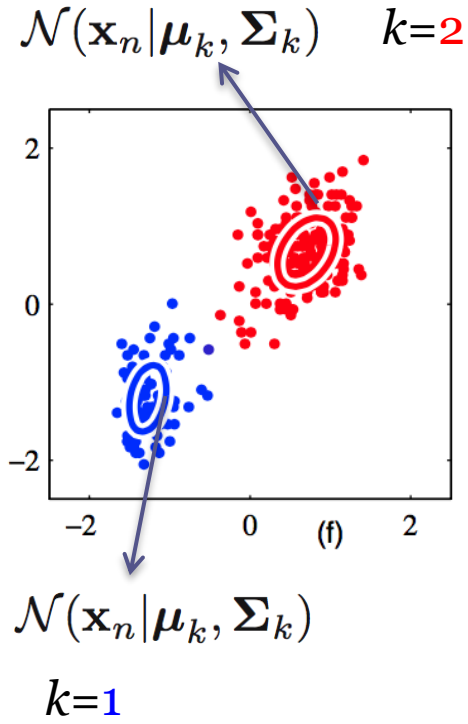
For all $\boldsymbol{m}_i, i = 1, \ldots, k$

$$\boldsymbol{m}_i \leftarrow \sum_t b_i^t \boldsymbol{x}^t / \sum_t b_i^t$$

Until $\boldsymbol{m}_i$ converge



# Gaussian Mixture Model (GMM)




We use $z_k = 1$ to indicate a point $\mathbf{x}$ belongs to cluster $k$

$$\mathbf{z} = (z_1, \ldots, z_K) \qquad z_k \in \{0, 1\} \qquad \sum_k z_k = 1$$

Assume the points in the same cluster follow a **Gaussian distribution**

$$p(\mathbf{x}|z_k = 1) = \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$$

A mixing weight for each cluster:

$$p(z_k = 1) = \pi_k \qquad 0 \leqslant \pi_k \leqslant 1 \qquad \sum_{k=1}^{K} \pi_k = 1$$

*prior probability of point belonging to a cluster*

So, we get a distribution for the data point $\mathbf{x}$:

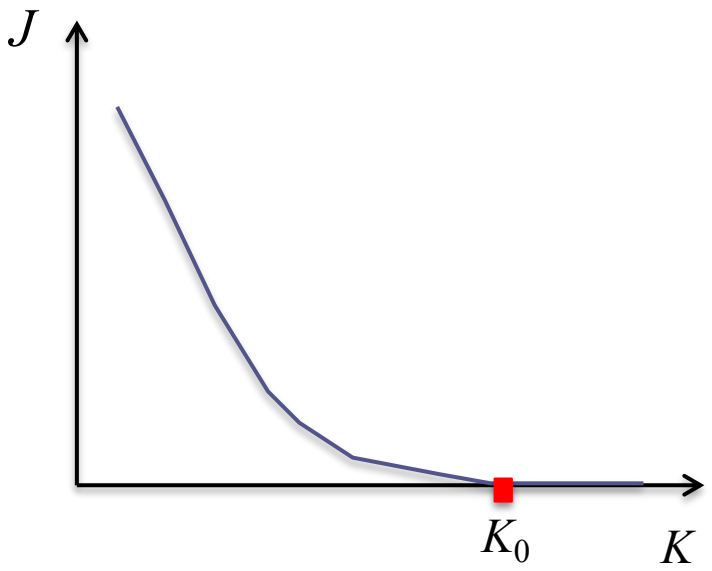
$$p(\mathbf{x}) = \sum_{\mathbf{z}} p(\mathbf{z}) p(\mathbf{x}|\mathbf{z}) = \sum_{k=1}^{K} \pi_k \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$$

# How to determine the cluster number K?

- This is a model selection problem.

## K-mean

$$J = \sum_{n=1}^{N}\sum_{k=1}^{K} r_{nk}\|\mathbf{x}_n - \boldsymbol{\mu}_k\|^2$$

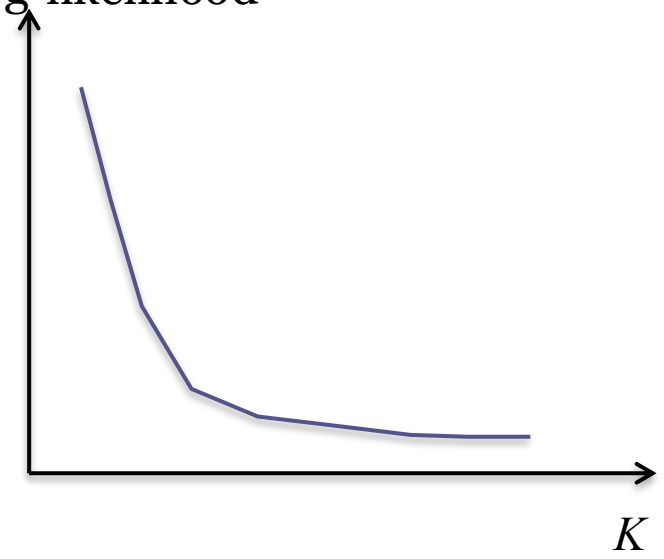



$J$ does not tell which $K$ is better.

## GMM

$$\ln p(\mathbf{X}|\boldsymbol{\pi},\boldsymbol{\mu},\boldsymbol{\Sigma}) = \sum_{n=1}^{N}\ln\left\{\sum_{k=1}^{K}\pi_k\mathcal{N}(\mathbf{x}_n|\boldsymbol{\mu}_k,\boldsymbol{\Sigma}_k)\right\}$$

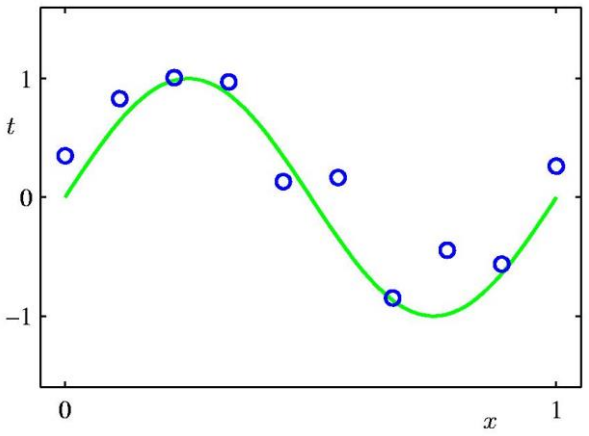
- Log-likelihood

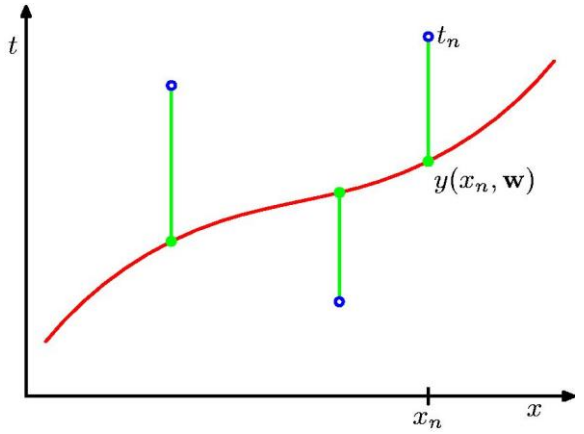


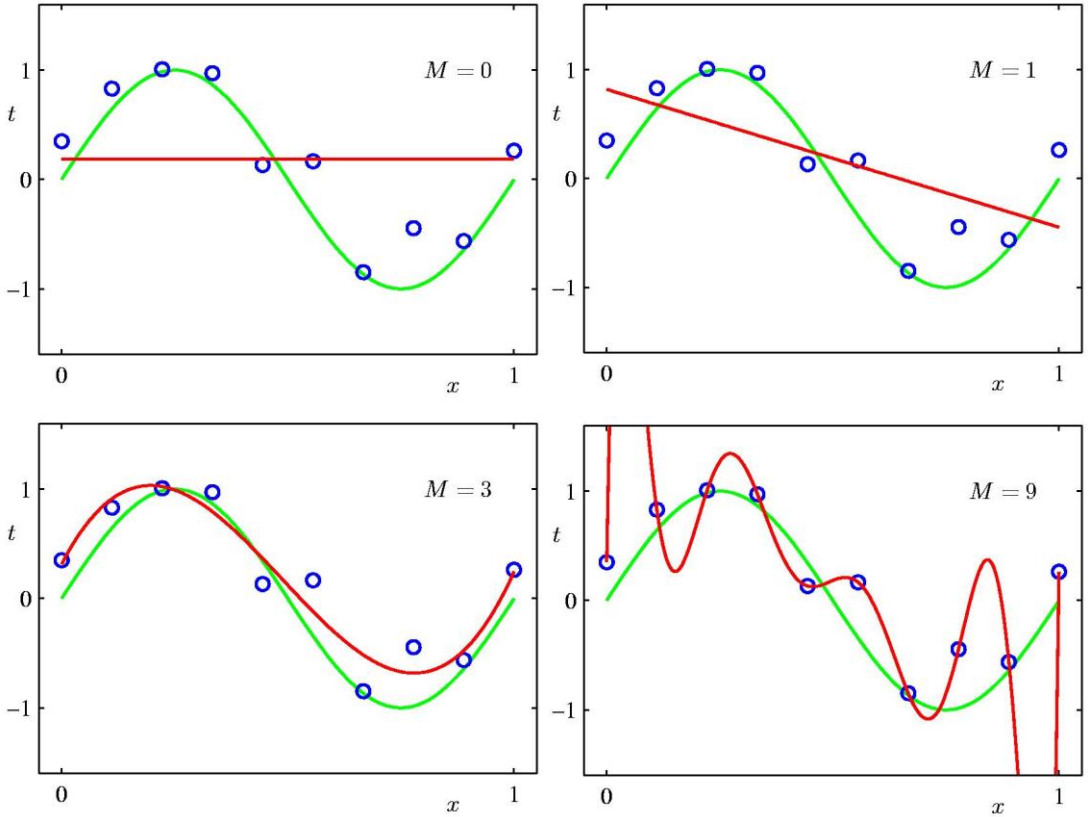

Negative log-likelihood also decreases as $K$ increases.

# Polynomial Curve Fitting

- Need to determine the coefficients and the degree

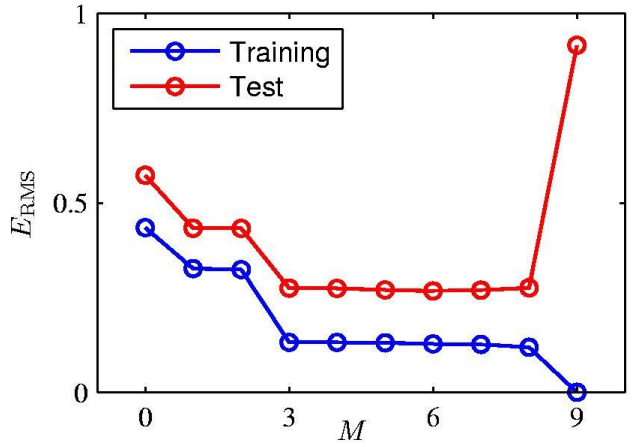



Mean Square Error (MSE)

$$y(x, \mathbf{w}) = w_0 + w_1 x + w_2 x^2 + \ldots + w_M x^M = \sum_{j=0}^{M} w_j x^j$$




Root-Mean-Square (RMS) Error: $E_{\mathrm{RMS}} = \sqrt{2E(\mathbf{w}^\star)/N}$

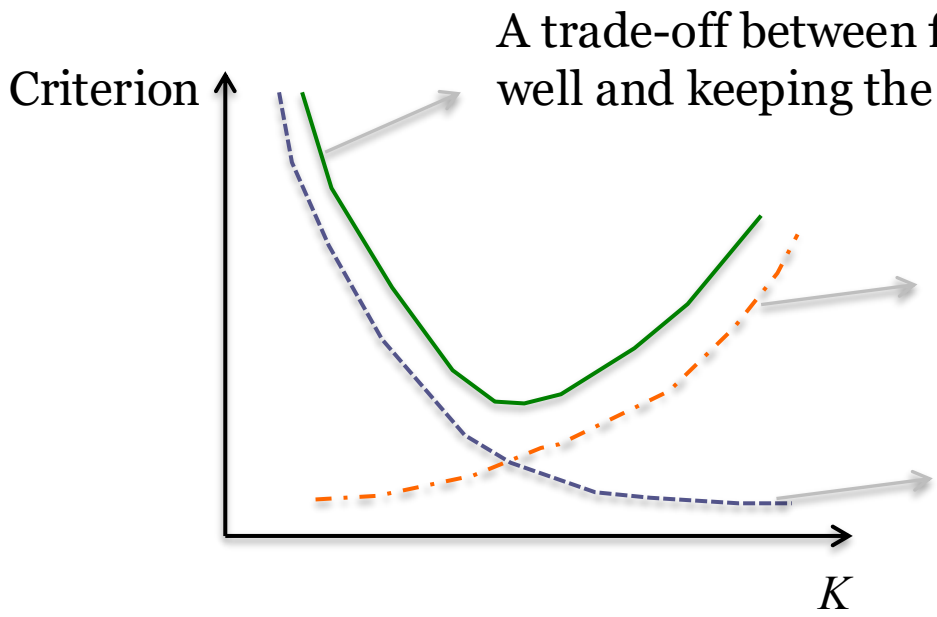
# Model selection for generalization

Probabilistic model

$$p(X_N \mid \Theta_K)$$

Candidate models:

$$\Theta_1 \subseteq \Theta_2 \subseteq \cdots \subseteq \Theta_K \subseteq \cdots$$




A trade-off between fitting the data well and keeping the model simple

Models become more and more complex as K increases.

Negative log-likelihood (or fitting error) to make sure the model fit the data well.

Akaike's Information Criterion (AIC) $\ln p(X_N \mid \hat{\Theta}_K) - d_k$

Bayesian Information Criterion (BIC) $\ln p(X_N \mid \hat{\Theta}_K) - \frac{1}{2} d_k \ln N$

$d_k$: number of free parameters
$N$: sample size

# Two-phase method for model selection

- Assume the optimal $K^*$ is within the range $[1, K_{\max}]$.

- Phase (1): For each $k = 1, \ldots, K_{\max}$, compute the maximum likelihood estimator:

$$\widehat{\Theta}_{ML}(k) = \underset{\Theta}{\operatorname{argmax}} \log[P(X|\Theta, k)]$$

- Phase (2): Select the optimal $K^*$ by optimizing the values of the model selection criterion J, e.g., **AIC**, **BIC**:

$$K^* = \underset{k}{\operatorname{argmax}} J(\widehat{\Theta}_{ML}(k))$$

Akaike's Information Criterion (AIC)

$$\ln p(X_N \mid \hat{\Theta}_K) - d_k$$

Bayesian Information Criterion (BIC)

$$\ln p(X_N \mid \hat{\Theta}_K) - \frac{1}{2} d_k \ln N$$

# 2. The history of MDL

# Inductive Inference (归纳推断)

- **Induction** is "*the process of inferring a general law or principle from the observations of particular instances*". (Oxford English Dictionary)

- **Science** is induction: *from observed data to physical laws*.

- But, how?

# Epicurus: Multiple Explanations

- **Epicurus** (342--270BC), a Greek philosopher of science, proposed the Principle of Multiple Explanations:

*If more than one theory is consistent with the observations, keep all theories*.

- "There are also some things for which it is not enough to state a single cause, but several, of which one, however, is the case. Just as if you were to see the lifeless corpse of a man lying far away, it would be fitting to state all the causes of death in order that the single cause of his death may be stated. For you would not be able to establish conclusively that he died by the sword or of cold or of illness or perhaps by poison, but we know that there is something of this kind that happened to him." [Lucretius]

# Occam's Razor



- Commonly attributed to William of Ockham (1290--1349).
- This was formulated about fifteen hundred years after Epicurus. In sharp contrast to the principle of multiple explanations, it states:

  ***Entities should not be multiplied beyond necessity.***

- Commonly explained as: *when have choices, choose the simplest theory*.
- Bertrand Russell: "It is vain to do with more what can be done with fewer."
- Newton (*Principia*): "Natura enim simplex est, et rerum causis superfluis non luxuriat". (自然的东西是简单的，而自然的东西是多余的，不是奢侈的)

# Kolmogorov complexity



- In algorithmic information theory, the Kolmogorov complexity of an object, such as a piece of text, is *the length of a shortest computer program* (in a predetermined programming language) that produces the object as output.

Consider the following two strings of 32 lowercase letters and digits:

`abababababababababababababababab` , and

`4c1j5b2p0cv4w1x8rx2y39umgw5q85s7`

```
function GenerateString1()
    return "ab" × 16
```

```
function GenerateString2()
    return "4c1j5b2p0cv4w1x8rx2y39umgw5q85s7"
```

# An Counter Example



- Once upon a time, there was a little girl named Emma. Emma had never eaten a banana, nor had she ever been on a train.
- One day she had to journey from New York to Pittsburgh by train. To relieve Emma's anxiety, her mother gave her a large bag of bananas.
- At Emma's first bite of her banana, the train plunged into a tunnel. At the second bite, the train broke into daylight again. At the third bite,  Lo! into a tunnel; the fourth bite, La! into daylight again.
- And so on all the way to Pittsburgh. Emma, being a bright little girl, told her grandpa at the station: ``Every odd bite of a banana makes you blind; every even bite puts things right again.'' (N.R. Hanson, Perception & Discovery)

# The paper by J. Rissanen in 1978

*Automatica*, Vol. 14, pp. 465–471
0005-1098/78/0901–0465 $02.00/0
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## Modeling By Shortest Data Description*

J. RISSANEN†

*Estimates of both integer-valued structure parameters and real-valued system parameters may be obtained from a model based on the shortest data description principle.*

**Key Word Index**—Modeling; parameter estimation; identification; statistics; stochastic systems.

**Summary**—The number of digits it takes to write down an observed sequence $x_1,\ldots,x_N$ of a time series depends on the model with its parameters that one assumes to have generated the observed data. Accordingly, by finding the model which minimizes the description length one obtains estimates of both the integer-valued structure parameters and the real-valued system parameters.

fusion because basically all the different entropies measure one or another aspect of description length, or to put it more bluntly, they all amount to the logarithm of the cardinality of a finite set or its limit. To read this paper no background in information theory is needed.

# History

- The **Minimum Description Length (MDL)** Principle has mainly been developed by J. Rissanen in a series of papers starting with [Rissanen 1978].
- It has its roots in the theory of **Kolmogorov or algorithmic complexity** [Li and Vitanyi 1997], developed in the 1960s by Solomonoff [1964], Kolmogorov [1965] and Chaitin [1966, 1969].
- Among these authors, Solomonoff (a former student of the famous philosopher of science, Rudolf Carnap) was explicitly interested in inductive inference. The 1964 paper contains explicit suggestions on how the underlying ideas could be made practical, thereby foreshadowing some of the later work on two-part MDL.
- While Rissanen was not aware of Solomonoff's work at the time, Kolmogorov's [1965] paper did serve as an inspiration for Rissanen's [1978] development of MDL.
- Another important inspiration for Rissanen was **Akaike's [1973] AIC** method for model selection, essentially the first model selection method based on information- theoretic ideas. Even though Rissanen was inspired by AIC, both the actual method and the underlying philosophy are substantially different from MDL.

# History

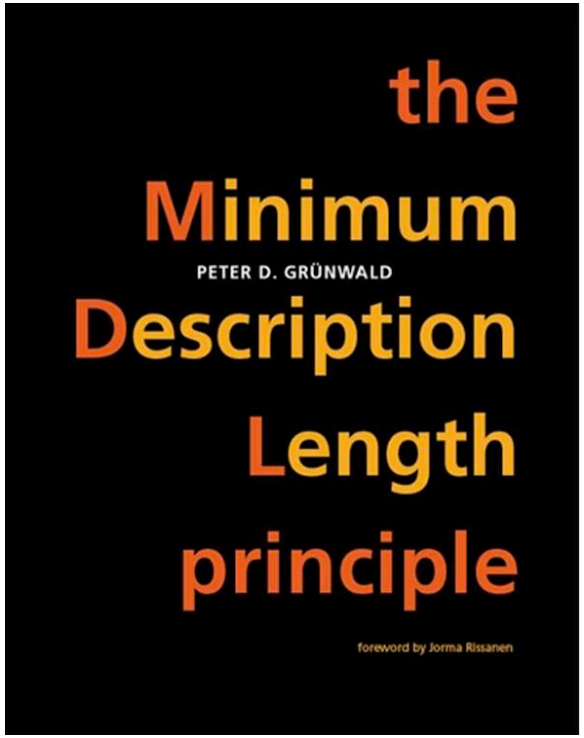
- MDL is much closer related to the **Minimum Message Length (MML)** Principle, developed by Wallace and his co-workers in a series of papers starting with the ground-breaking [Wallace and Boulton 1968].
- Other milestones are [Wallace and Boulton 1975] and [Wallace and Freeman 1987].
- Remarkably, Wallace developed his ideas without being aware of the notion of Kolmogorov complexity.
- Although Rissanen became aware of Wallace's work before the publication of [Rissanen 1978], he developed his ideas mostly independently, being influenced rather by Akaike and Kolmogorov.
- Indeed, despite the close resemblance of both methods in practice, the underlying philosophy is quite different.
- The first publications on MDL only mention **two-part codes**. Important progress was made by Rissanen [1984], in which prequential codes are employed for the first time and [Rissanen 1987], introducing the Bayesian mixture codes into MDL.
- This led to the development of the notion of **stochastic complexity** as the shortest codelength of the data given a model [Rissanen 1986; Rissanen 1987].

# History

- However, the connection to Shtarkov's **normalized maximum likelihood code** was not made until 1996, and this prevented the full development of the notion of 'parametric complexity'.
- In the mean time, in his impressive Ph.D. thesis, Barron [1985] showed how a specific version of the two-part code criterion has excellent frequentist statistical consistency properties.
- This was extended by Barron and Cover [1991] who achieved a breakthrough for two-part codes: they gave clear prescriptions on how to design codes for hypotheses, relating codes with good minimax codelength properties to rates of convergence in statistical consistency theorems.
- Some of the ideas of Rissanen [1987] and Barron and Cover [1991] were, as it were, unified when Rissanen [1996] introduced a new definition of **stochastic complexity** based on the normalized maximum likelihood code.
- The resulting theory was summarized for the first time by Barron, Rissanen, and Yu [1998], and is called 'refined MDL' in the present overview.

# 3. A Tutorial on MDL

## 3.1 An informal overview




Peter D. Grünwald, The Minimum Description Length Principle, MIT Press, March 23, 2007

# MDL is based on the following insights

- The Minimum Description Length (MDL) Principle is a relatively recent method for **inductive inference** that provides a generic solution to the **model selection** problem.
- Any **regularity** in the data can be used to **compress the data**, i.e., to describe it using fewer symbols than the number of symbols needed to describe the data literally.
- The more regularities there are, the more the data can be compressed.
- **Equating 'learning' with 'finding regularity'**, we can therefore say that the more we are able to compress the data, the more we have learned about the data.
- Formalizing this idea leads to a general theory of inductive inference with several attractive properties: 1) Occam's Razor; 2) No overfitting, automatically; 3) Bayesian interpretation; 4) No need for "underlying truth"; 5) Predictive interpretation.

# Properties of MDL

- **Occam's Razor**
  - MDL chooses a model that **trades-off** *goodness-of-fit* on the observed data with '*complexity*' *or* '*richness*' *of the model.* As such, MDL embodies a form of Occam's Razor, a principle that is both intuitively appealing and informally applied throughout all the sciences.
- **No overfitting, automatically**
  - MDL procedures automatically and inherently protect against overfitting and can be used **to estimate both the parameters and the structure (e.g., number of parameters) of a model**. In contrast, to avoid over- fitting when estimating the structure of a model, traditional methods such as maximum likelihood must be modified and extended with additional, typically ad hoc principles.
- **Bayesian interpretation**
  - MDL is closely related to Bayesian inference, but avoids some of the interpretation difficulties of the Bayesian approach1, especially in the realistic case when it is known a priori to the modeler that none of the models under consideration is true.

# Properties of MDL

- **No need for ‘underlying truth’**
  - In contrast to other statistical methods, MDL procedures have a clear interpretation independent of whether or not there exists some underlying ‘true’ model.
- **No overfitting, automatically**
  - Because data compression is formally equivalent to a form of probabilistic prediction, MDL methods can be interpreted as searching for a model with good predictive performance on unseen data.

# The Fundamental Idea: Learning as Data Compression

- **Regularity**

Consider the following three sequences. We assume that each sequence is 10000 bits long, and we just list the beginning and the end of each sequence.

$$00010001000100010001 \ \ldots \ 00010001000100010001000100010001 \quad (1.1)$$

$$01110100110100100110 \ \ldots \ 10101110101110110001011000010 \quad (1.2)$$

$$00011000001010100000 \ \ldots \ 00100010000100000010001100000 \quad (1.3)$$

1. The first sequence is a 2500-fold repetition of 0001. Intuitively, the sequence looks regular; there seems to be a simple 'law' underlying it.
2. The second sequence has been generated by tosses of a fair coin. It is intuitively speaking as 'random as possible', and in this sense there is no regularity underlying it.
3. The third sequence contains approximately four times as many 0s as 1s. It looks less regular, more random than the first; but it looks less random than the second. There is still some discernible regularity in these data, but of a statistical rather than of a deterministic kind.

# The Fundamental Idea: Learning as Data Compression

- **Compression**
  - Any regularity detected in the data can be used to compress the data, i.e., to describe it in a short manner.
  - Descriptions are always relative to some *description method* which maps descriptions $D'$ in a unique manner to data sets $D$.
  - A particularly versatile description method is a general-purpose computer language like C or Pascal.
  - A description of $D$ is then any computer program that prints $D$ and then halts.

1. Using a language similar to Pascal, we can write a program to print the first sequence, but is clearly a lot shorter.

```
for i = 1 to 2500; print '0001'; next; halt
```

2. if one generates a sequence like the previous 2nd sequence by tosses of a fair coin, then with extremely high probability, the shortest program that prints it and then halts will look something like this:

```
print '01110100110100001010101.....10101110101110110001011000010'; halt
```

3. If The third sequence lies in between the first two, and it can be compressed to some length $\alpha n$, with $0 < \alpha < 1$.

# Examples: compressing various regular sequences

- **The Number π**
  - Evidently, there exists a computer program for generating the first $n$ digits of π – such a program could be based, for example, on an infinite series expansion of π.
  - This computer program has constant size, except for the specification of $n$ which takes no more than $O(\log n)$ bits.
  - Thus, when $n$ is very large, the size of the program generating the first $n$ digits of π will be very small compared to $n$.
  - The π-digit sequence is deterministic, and therefore extremely regular.

# Examples: compressing various regular sequences

- **Natural Language**
  - Most sequences of words are not valid sentences according to the English language.
  - This fact can be exploited to substantially compress English text, as long as it is syntactically mostly correct:
  - by first describing a grammar for English, and then describing an English text *D* with the help of that grammar, *D* can be described using much less bits than are needed without the assumption that word order is constrained.

# Kolmogorov Complexity and Ideal MDL

- The Kolmogorov complexity of an object, such as a piece of text, is the length of a shortest computer program (in a predetermined programming language) that produces the object as output.
- The lower the Kolmogorov complexity of a sequence, the more regular it is.
- **Invariance theorem**: As long as the sequence *D* is long enough, it is not essential which computer language one chooses, as long as it is general-purpose (such as C or Pascal).
- Solomonoff's paper, called *A Theory of Inductive Inference*, contained the idea that the ultimate model for a sequence of data may be identified with the shortest program that prints the data. Solomonoff's ideas were later extended by several authors, leading to an 'idealized' version of MDL.

# Kolmogorov Complexity and Ideal MDL

- This idealized MDL is very general in scope, but not practically applicable, for the following two reasons:

1. **Uncomputability**
   - It can be shown that there exists no computer program that, for every set of data *D*, when given *D* as input, returns the shortest program that prints D [Li and Vit´anyi 1997].
2. **Arbitrariness/dependence on syntax**
   - In practice we are confronted with small data samples for which the invariance theorem does not say much. Then the hypothesis chosen by idealized MDL may depend on arbitrary details of the syntax of the programming language under consideration.

# MDL: The Basic Idea

- The goal of statistical inference may be cast as trying to **find regularity** in the data.
- Regularity may be identified with 'ability to compress'.
- MDL combines these two insights by *viewing learning as data compression*.
- It tells us that, for a given set of hypotheses *H* and data set *D*, we should try to find the hypothesis or combination of hypotheses in H that compresses *D* most.
- This idea can be applied to all sorts of inductive inference problems, but it turns out to be most fruitful in (and its development has mostly concentrated on) problems of *model selection* and, more generally, dealing with *overfitting*.

# Example (model selection and overfitting)

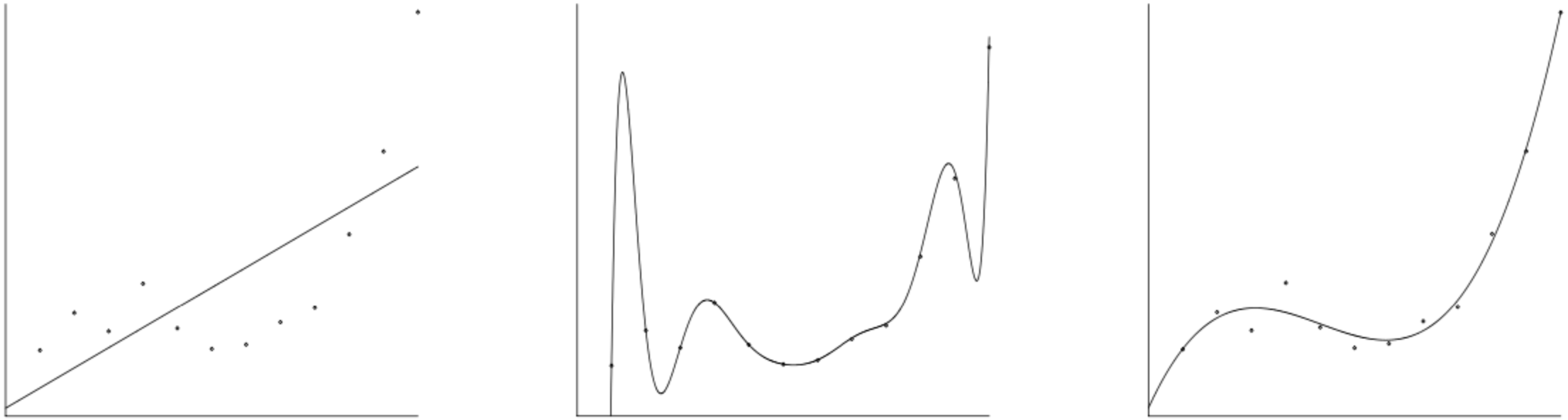


Figure 1.1: A simple, a complex and a trade-off (3rd degree) polynomial.

# Models vs. Hypotheses

- We use the phrase ***point hypothesis*** to refer to a single probability distribution or function.
  - An example is the polynomial $5x^2 + 4x + 3$.
  - A point hypothesis is also known as a 'simple hypothesis' in the statistical literature.
- We use the word ***model*** to refer to a family (set) of probability distributions or functions with the same functional form.
  - An example is the set of all second-degree polynomials.
  - A model is also known as a 'composite hypothesis' in the statistical literature.
- We use *hypothesis* as a generic term, referring to **both** point hypotheses and models.

# Example

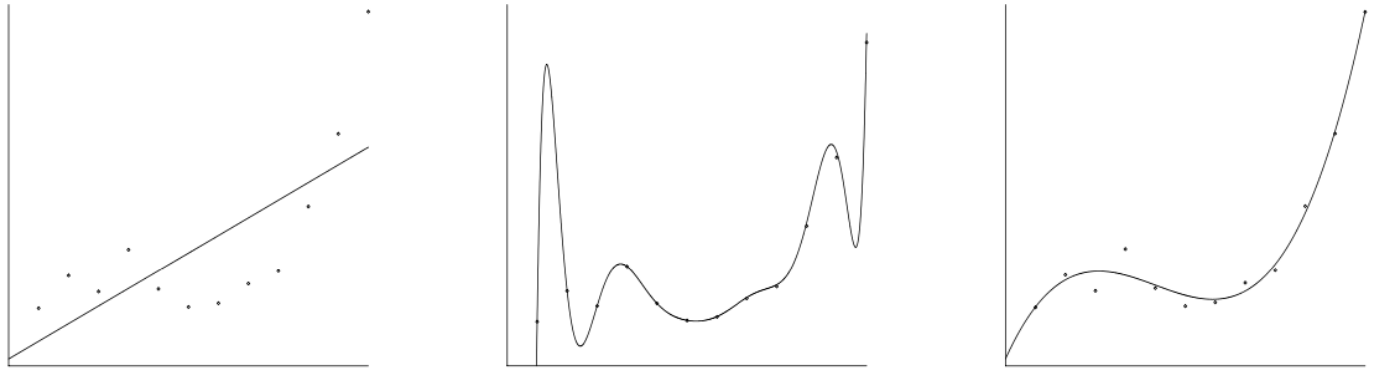


Figure 1.1: A simple, a complex and a trade-off (3rd degree) polynomial.

- In this terminology, considering the problem described in the above:
  - It is a '**hypothesis selection problem**' if we are interested in selecting **both** the degree of a polynomial and the corresponding parameters;
  - It is a '**model selection problem**' if we are mainly interested in selecting the degree.

# The two-part code version of MDL

Let $\mathcal{H}^{(1)}, \mathcal{H}^{(2)}, \ldots$ be a list of candidate models (e.g., $\mathcal{H}^{(k)}$ is the set of $k$-th degree polynomials), each containing a set of point hypotheses (e.g., individual polynomials). The best point hypothesis $H \in \mathcal{H}^{(1)} \cup \mathcal{H}^{(2)} \cup \ldots$ to explain the data $D$ is the one which minimizes the sum $L(H) + L(D|H)$, where

- $L(H)$ is the length, in bits, of the description of the hypothesis; and
- $L(D|H)$ is the length, in bits, of the description of the data when encoded with the help of the hypothesis.

The best *model* to explain $D$ is the smallest model containing the selected $H$.

# Example (selecting polynomials)

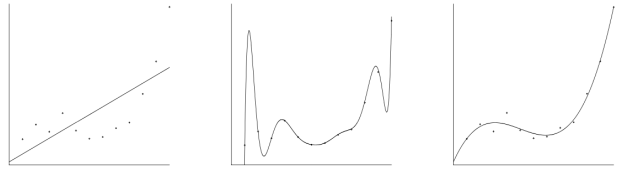


Figure 1.1: A simple, a complex and a trade-off (3rd degree) polynomial.

- In our previous example, the candidate hypotheses were polynomials.
- We can describe a polynomial by describing its coefficients in a certain precision (number of bits per parameter). Thus, *the higher the degree of a polynomial or the precision, the more bits we need to describe it and the more 'complex' it becomes*.
- A description of the data 'with the help of' a hypothesis means that *the better the hypothesis fits the data, the shorter the description will be*.
- We only have to code the *errors* the hypothesis makes on the data rather than the full data. In our polynomial example, the better a polynomial $H$ fits $D$, the fewer bits we need to encode the discrepancies between the actual $y$-values $y_i$ and the predicted y-values $H(x_i)$.
- We can typically find a very complex point hypothesis (large L(H)) with a very good fit (small L(D|H)).
- We can also typically find a very simple point hypothesis (small L(H)) with a rather bad fit (large L(D|H)).
- The sum of the two description lengths will be minimized at a hypothesis that is quite (but not too) 'simple', with a good (but not perfect) fit.

# Definition of L(D|H)

- Assume that all $H$ under consideration define probability distributions.
- If $H$ is a polynomial, we can turn it into a distribution by making the additional assumption that the $Y$-values are given by $Y = H(X) + Z$, where $Z$ is a normally distributed noise term.
- For each $H$, we need to define a code with length $L(\cdot \mid H)$ such that $L(D|H)$ can be interpreted as 'the codelength of $D$ when encoded with the help of $H$'.
- It turns out that for probabilistic hypotheses, there is only one reasonable choice for this code. It is the so-called Shannon-Fano code, satisfying, for all data sequences $D$, $L(D|H) = -\log P(D|H)$, where $P(D|H)$ is the probability mass or density of $D$ according to $H$ – such a code always exists.

# Definition of L(H): A Problem for Crude MDL

- It is more problematic to find a good code for hypotheses $H$
- Since the **description length $L(H)$** of any fixed point hypothesis $H$ **can be very large under one code**, but quite **short under another**, the procedure is in danger of becoming arbitrary.
- In the first publications on MDL [Rissanen 1978; Rissanen 1983], it was advocated to choose some sort of *minimax code* for $H$, minimizing, in some precisely defined sense, the **shortest worst-case total description length** $L(H) + L(D|H)$, where the worst-case is over all possible data sequences.
- Thus, the MDL Principle is employed at a 'meta-level' to choose a code for $H$. However, this code **requires a cumbersome discretization of the model space** $H$, which is not always feasible in practice.

# Refined MDL

- Rissanen [1984] realized that these problems could be side-stepped by using a *one-part* rather than a *two-part code*.
- This development culminated in 1996 in a completely precise prescription of MDL for many, but certainly not all practical situations [Rissanen 1996]. We call this modern version of MDL as **refined MDL**.
- In refined MDL, we associate a code for encoding $D$ **not with a single** $H \in \mathcal{H}$, but with the full model $H$. Thus, given model H, we encode data not in two parts but we design a single **one-part code** with lengths $\bar{L}(D|\mathcal{H})$.
- This code is designed such that *whenever there is a member of (parameter in) H that fits the data well, in the sense that* $L(D|H)$ *is small, then the codelength* $\bar{L}(D|\mathcal{H})$ *will also be small.*

# Refined MDL (the stochastic complexity)

- Thus, given model H, we encode data not in two parts but we design a single **one-part code** with lengths $\bar{L}(D|\mathcal{H})$.
- This code is designed such that *whenever there is a member of (parameter in) H that fits the data well, in the sense that $L(D|H)$ is small, then the codelength $\bar{L}(D|\mathcal{H})$ will also be small.*
- Codes with this property are called **universal codes** in the information-theoretic literature [Barron, Rissanen, and Yu 1998].
- Among all such universal codes, we pick the one that is *minimax optimal*.
- For example, the set $\mathcal{H}^{(3)}$ of third-degree polynomials is associated with a code with lengths $\bar{L}(D|\mathcal{H}^{(3)})$ such that, the better the data $D$ are fit by the best-fitting third-degree polynomial, the shorter the codelength $\bar{L}(D|\mathcal{H}^{(3)})$.
- $\bar{L}(D|\mathcal{H})$ is called the **stochastic complexity** of the data given the model.

# Parametric Complexity

- The **parametric complexity** of a parametric model $\mathcal{H}$ is denoted by **COMP**($\mathcal{H}$). This is *a measure of the richness of model $\mathcal{H}$, indicating its ability to fit random data.*
- This complexity is related to the **degrees-of-freedom** in $\mathcal{H}$, but also to the **geometrical structure** of $\mathcal{H}$. $y = ax^b + Z$ $y = a\ln(x+b) + Z$
- For given data $D$, $\widehat{H}$ denote the distribution in $\mathcal{H}$ which maximizes the probability, and hence minimizes the codelength $L(D|\widehat{H})$ of $D$. It turns out that

$$\text{stochastic complexity of } D \text{ given } \mathcal{H} = L(D \mid \hat{H}) + \mathbf{COMP}(\mathcal{H}).$$

- Therefore, the refined MDL model selection still involves a trade-off between two terms: a goodness-of-fit term $L(D|\widehat{H})$ and a complexity term **COMP**($\mathcal{H}$).
- However, because we do not explicitly encode hypotheses $H$ any more, there is no arbitrariness any more.

# Example

- Refined MDL allows us to compare models of different functional form. It even accounts for the phenomenon that *different models with the same number of parameters may not be equally complex*.
- Consider two models from psychophysics describing the relationship between physical dimensions (e.g., light intensity) and their psychological counterparts (e.g. brightness). [Myung, Balasubramanian, and Pitt 2000]

**Stevens' model** $y = ax^b + Z$ versus $y = a\ln(x+b) + Z$ **Fechner's model**

where $Z$ is a normally distributed noise term.

- Both models have two free parameters. But, Stevens' model is more flexible or complex than Fechner's.
- Myung et al. [2000] generated many samples of size 4 from Fechner's model, using some fixed parameter values. They then fitted both models to each sample. In 67% of the trials, Stevens' model fitted the data better than Fechner's, even though the latter generated the data.
- In refined MDL, the 'complexity' associated with Stevens' model is much larger than the complexity associated with Fechner's, and if both models fit the data equally well, MDL will prefer Fechner's model.

# The MDL philosophy

- The **first** central MDL idea is that *every regularity in data may be used to compress that data*.
- The **second** central idea is that *learning can be equated with finding regularities in data.*
- It implies that *methods for learning from data must have a clear interpretation independent of whether any of the models under consideration is 'true' or not.*

> "We never want to make the false assumption that the observed data actually were generated by a distribution of some kind, say Gaussian, and then go on to analyze the consequences and make further deductions. Our deductions may be entertaining but quite irrelevant to the task at hand, namely, to learn useful properties from the data."
>
> *Jorma Rissanen [1989]*

# Thank you!